Demand Planning UI: how it works and glossary

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Purpose

This tool predicts how many Eveready products will sell each week and each month, and checks the sales team's forecasts against budget, last year and stock. It runs on made-up (mock) data and was built by AnvIQ Labs for the CFO pitch.

The screens follow the Demand Planning section of the CFO's expectations workbook:

CFO item	What he asked for	Where it lives in the UI
1. Forecast accuracy	Rolling accuracy at M3, M2 and M1 against industry and Eveready's own ~68% at M3, with over-forecast and under-forecast shown separately	Forecast screen: accuracy card and the Monthly runs tab
2. Forecast challenge	A flag on every sales forecast that is out of line with budget, last year or depot stock	Challenge screen
3. New product forecasting	Demand for products with no history, using competitor (Nielsen-style) data	Not built yet
4. New channels (quick commerce)	Forecast demand without partners over-ordering about 10x, and measure OTIF	Quick commerce screen
5. Weather-driven seasonality	Week-level shocks and their recovery, not only seasonal averages	Weather screen

Data behind the screens

What we forecast: the units Eveready sells to its distributors each week (called primary sales), for each product, for the whole country and for each Kolkata depot. The CFO's workbook does not say which sales level he means, so this is our working choice.

The pipeline reads three files:

File	What it holds	Used by
`eveready_mock_data_v2.csv` (91 columns, 26,934 rows)	Weekly sales, sales-team forecasts at 3, 2 and 1 months ahead, budget, orders, stock, quick commerce, lost sales, transfers, MOQ columns	Every screen
`ext.csv` (external factors)	Weekly weather, power cuts, heatwave and cyclone flags, dengue, zinc price, rural income, LED price, expected values, festival labels	Forecast features, Weather screen, driver panels
`competitor.csv` (Nielsen-style)	Retail sales, distribution and promotion for 35 competitor brands, including 8 liquid vaporizer brands	Not used by any screen yet; reserved for new product forecasting

Coverage: 157 weeks (Oct 2023 to Sep 2026), 30 products, the whole country, and five Kolkata depots (Barasat, Barrackpore, Baruipur, Dankuni, Taratala). The depots are only about 7% of national sales, so depot screens describe Kolkata, not India.

Some columns are blank on purpose. For example, quick-commerce columns are filled only for the 16 products sold there, and four new products start selling later than the rest.

Earlier versions of the screens used an edited copy of this file. They now use the file exactly as supplied.

The screens

Four screens are built. The left menu also lists Overview, New products, Supply & inventory, Channels & events and Guide, shown greyed out as "Not built yet". The address in the browser holds the screen and tab (for example `#/forecast?view=monthly`), so refresh, back and shared links return to the same place.

Controls shared by every screen

Control	Options	Notes
Category, Product	All, or one category or product	Changing the category resets the product
Level	National, Kolkata depots	Quick commerce and Weather are national only
Measure	Units, Value (₹)	Hidden on the Weather screen
Light / dark	Toggle in the top bar	Charts re-read their colours on change
Scenario controls	Button on the Forecast screen	Described below

Forecast & accuracy

The weekly outlook is the main view. It shows every week up to March 2027: actuals up to the last week of data, then the model forecast only from 5 Oct 2026. Headline tiles sit above a chart with these lines: Actual, Forecast (ensemble), P50 (median), Sales-team forecast (3 months ahead), Last year, Budget, the Model M3, M2 and M1 runs, and Supply-constrained (it appears only when supply availability is set below 100%). Budget and the M3, M2 and M1 runs are hidden when the screen opens; click a legend entry to show them. A Table view lists the same numbers by week.

Clicking a week opens a panel that explains it: what each model says (seasonal naive, ETS and LightGBM, each as a bar showing how far it pulls the forecast from the starting point, times its weight in the blend), what was going on that week, and how the week compares with last month or the same month last year. When ETS is blank for a week, the panel lists only the models that are active.

The Monthly runs tab shows each target month as the M3, M2 and M1 runs made it, against actuals and the sales team, with a Target month selector. Under it, the accuracy card scores the runs against the industry reference and Eveready's own figure (see How accuracy is measured). The monthly tab does not apply scenario controls.

Scenario controls

A side panel with three levers and four presets. Nothing changes on the page until Apply is pressed (draft, then apply), and Reset returns to the base case.

Lever	Range	What it does
Overall demand	−20% to +20%	Moves every forecast week by the same percentage
Festive weeks	−30% to +30%	An extra change in weeks that fall in a festival or its sell-in period
Supply availability	80% to 100%	Share of demand the plants can ship; draws a separate dotted line and leaves demand unchanged

Presets: Base case, Weak festive season (−3% overall, −15% festive), Strong festive season (+2% overall, +15% festive) and Supply squeeze (88% availability). The levers scale the forecast weeks of the Forecast and P50 lines. The models are not re-run, so a scenario is an overlay and not a new forecast.

Forecast challenge

Every sales forecast for a chosen target month and lead (M3, M2 or M1) is checked against four references: budget, last year, the model, and current depot stock. A forecast is flagged when enough checks fail. The rules are adjustable: budget band (±20% to ±60%), last-year band (±25% to ±100%), model band (±25% to ±75%), how many of the three reference checks must fail (1 to 3), and maximum stock cover (6 to 16 weeks). The table can show Flagged forecasts only or All, and a note counts the flags raised by type.

Quick commerce & OTIF

National only. It compares what quick-commerce partners ordered against what they actually sold to shoppers (their sell-out), shows delivery performance (OTIF against fulfilment, which differ because OTIF scores whole order lines), and plans the next 13 weeks of shipments given the stock partners hold. Controls: a window of 13 weeks, 26 weeks or all, and the buffer stock a partner keeps (1, 2 or 3 weeks). A By product table closes the screen.

Weather & shocks

National only. For a chosen monsoon season (2024, 2025 or 2026), it shows demand against the forecast made just before the weather event, and actual rain against expected rain, as a Chart or a Table, so a shock and its recovery can be read week by week.

How the forecast is made

The forecast blends three models. Each one predicts every product's sales for each week, up to 26 weeks ahead, and the blend averages them using fixed weights. If one model has no value for a week, the others share its weight.

Model	Weight (country)	Weight (depots)	In plain words
Seasonal naive	20%	30%	Same week last year, adjusted for how sales are running now
Damped ETS	30%	0%	Smooths the sales history to find the level, trend and yearly pattern; needs two years of history, so it runs nationally only
LightGBM	50%	70%	A machine-learning model that learns from recent sales, last year, the calendar, price, weather and festivals

Festivals. Durga Puja and Diwali fall on different dates each year, so "the same week last year" often misses them. Around festivals, the models compare with last year's festival week instead, and LightGBM is told how many weeks away the nearest festival is.

Training. LightGBM learns from past sales. It predicts the change from a recent level, and odd single weeks are given less weight so they do not distort it.

M3, M2 and M1. At each month-end the models are retrained on everything known so far. The forecast made 3, 2 or 1 months before a month is that month's M3, M2 or M1 forecast.

P50. The median forecast: half of outcomes are expected above it and half below. It is set from how far off the model was in past tests.

Scenarios are applied afterwards, as a percentage overlay on the forecast weeks. They do not retrain anything.

How accuracy is measured and read

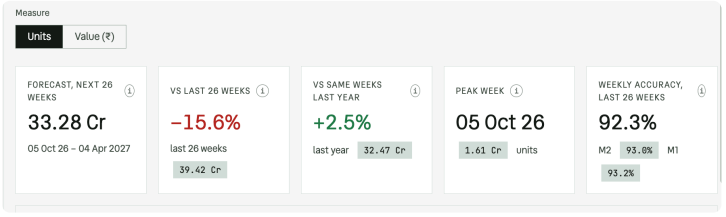
Accuracy is 1 minus (total forecast error divided by total actual sales), so 90% means the forecast was off by 10% of sales. The accuracy card adds all products together, then scores the M3, M2 and M1 forecasts over the ten completed months (Dec 2025 to Sep 2026). It also shows over-forecasts and under-forecasts separately, as the CFO asked, because they can cancel each other out inside a total.

Three reference points sit on the card: the industry reference (about 80% at M3 rising to 90% at M1, with M2 taken as the midpoint, 85%), Eveready's own figure quoted by the CFO (about 68% at M3), and the sales team's own forecast, scored the same way.

Weeks into months. A week can straddle two months, so each week is split by days. A week with 4 days in one month and 3 in the next counts 4/7 and 3/7.

Results on the 91-column v2 data (all products added together, ten months):

Level	Run	Model	Sales team
National	M3	93.9%	85.8%
National	M2	91.6%	89.1%
National	M1	92.7%	93.3%
Kolkata depots	M3	87.3%	88.1%
Kolkata depots	M2	89.2%	88.5%
Kolkata depots	M1	92.4%	88.7%



Scored product by product and month by month, the model is below the sales team nationally (64.7%, 69.4% and 78.3% against 72.3%, 75.1% and 83.9%) and about level at depot level. The pooled total flatters the model because errors in different products cancel. The sales team's weekly product-level accuracy in this data is 67%, 67.5% and 77% at M3, M2 and M1 (national, weekly).

How the data is rebuilt

The screens read ready-made data files. Scripts in the project's pipeline folder (`api/pipeline/`) produce them, run in this order:

1. `run_clean.py` for National and for Kolkata depot: backtests the ensemble from week 104 and makes the final forecast (about 3 minutes nationally, 2 for depots).
2. `run_band.py` for each level: measures backtest errors at 13, 19 and 26 weeks ahead, used for the P50 line and the range (about 2.5 and 1.8 minutes).

3. `build_vintages.py` for each level: retrains at each month-end and writes the M3, M2 and M1 runs (about 1.6 and 4 minutes). The two levels are then merged into `vintages.json`, and `add_hist.py` adds monthly actuals back to Nov 2023.
4. `build_weather.py`, `build_ui_data.py`, `build_challenge.py`, `build_qc.py` and `build_drivers.py`: the page data for each screen.
5. Copy `ui_data.json`, `vintages.json`, `challenge.json`, `weather.json`, `qc.json` and `drivers.json` into `api/data/`.

The older `run_all.sh` script is out of date and cannot rebuild everything on its own.

Glossary

Term	Meaning
Accuracy	One minus total absolute error divided by total actual sales, so 90% means the forecast was off by 10% of sales in total
Actual demand	Primary sales plus lost sales: what customers wanted, not only what shipped (an idea discussed, not yet a column)
Anchor week	The week of Durga Puja or Diwali in a given year, used to line up festival effects across years
Backtest	Re-running the models on past weeks as if the future were unknown, then scoring them against what happened
Baseline (`lbase`)	The recent sales level that LightGBM predicts a change from
Budget	The bottom-up sales budget in units and value for each week and product
Bottom-up	Built from product and depot detail upward, not set as one top-down number
Category	A product group such as LED bulbs, Batteries — alkaline or Decorative & festive; 10 in the data
Challenge (forecast challenge)	Checking a sales or supply forecast against budget, last year, the model and stock, and flagging what is out of line

Term	Meaning
Cold start	A new listing or channel with no sales history, such as a new quick-commerce product
Consensus forecast	The agreed one-month-ahead forecast column in the data
Cover (stock cover)	How many weeks current stock would last at recent sales
Cr (crore)	10 million; the UI shows units and rupees in crore
Damped ETS	Exponential smoothing with a trend that fades out over time; one of the three models
Dark store	A small warehouse used by quick-commerce platforms; the data has no dark-store split
Delayed monsoon	A weather event where rains arrive late and demand for some products swings by 50–60%, then recovers
Depot	A regional stock point; five in Kolkata (Barasat, Barrackpore, Baruipur, Dankuni, Taratala)
Depot imbalance	Some depots overstocked while others are short, with transfers recommended between them
Distributor stock	Stock held by distributors; rises by primary sales minus secondary sales each week
Diwali (Kali Puja)	The autumn festival of lights; a major sell-in period for lighting products
Durga Puja	The main West Bengal festival, in Sep–Oct; the biggest sell-in period
Ensemble	The blend of seasonal naive, ETS and LightGBM that makes the forecast
ETS	Exponential smoothing: a statistical model of level, trend and seasonality
External factors	Weather, power cuts, dengue, zinc price, rural income, LED price and festival flags that the models can use
Festive stretch	From 6 weeks before to 8 weeks after the Durga Puja week

Term	Meaning
Festival-aligned seasonal naive	A seasonal naive that compares festive weeks with last year's festival week, not the same calendar week
Fill rate (fulfilment)	Units delivered divided by units ordered; about 96% for quick commerce in the data
Forecast override reason	Why the sales team changed a forecast: distributor feedback, festival sell-in, weather, scheme loading, supply constraint, competitor activity or NPD launch plan
Gain	A measure of how much a feature improves LightGBM's splits; used to rank feature importance
Horizon	How many weeks ahead a forecast looks; 1 to 26
Huber loss	A training rule that treats large errors more gently than small ones, so single odd weeks matter less
IBP (SAP IBP)	SAP's integrated business planning suite; the CFO wants something simpler
Industry reference	About 80% accuracy at M3 rising to about 90% at M1
Kojagari Lakshmi Puja	The full-moon puja after Durga Puja, a lighting-related festival week
Lead (M3, M2, M1)	How many months before the target month a forecast was made
LightGBM	A gradient-boosted tree model; half of the national blend
Lost sales	Orders placed but not shipped (orders received minus primary sales, never below zero); it does not include demand that never became an order
M3 / M2 / M1	The forecast made 3, 2 or 1 month(s) before the target month
MOQ	Minimum order quantity set by a supplier
Mock data	Synthetic data standing in for Eveready's real records

Term	Meaning
NPD	New product development: forecasting products with no sales history
Nielsen	A retail audit provider; the competitor file imitates its data
Numeric distribution	Share of shops that stock a product
OTIF	On time, in full: the share of order lines delivered both on time and complete (84% against 96% fill in the data)
Over-forecast	A forecast above actual sales; also a sales-loss signal, because stock is tied up in the wrong place
Over-ordering	Quick-commerce partners ordering far more than they sell: about 10x in a product's first weeks, settling near 2x
P1 / P2	Priority classes for B2B orders; P1 is protected first when stock or capacity is short, with B2C cut first
P50	The median forecast line: half of outcomes expected above, half below
Partner sell-out	What a quick-commerce partner sells to shoppers
Price break	A supplier discount for buying above a set quantity; a second, deeper break sits at twice that
Primary sales	What Eveready sells to its distributors; the forecast target
Pro-rating	Splitting a week across the months its days fall in (days divided by 7)
Quick commerce	Fast-delivery retail platforms (dark stores); 16 listed products in the data
Retrain	Re-fitting the models on newer data; done at each month-end for the monthly runs
Safety stock	Extra stock held to cover surprises in demand or supply
Scenario controls	Overlays for overall demand, festive weeks and supply availability on the forecast weeks

Term	Meaning
Seasonal naive	Forecasts a week as the same week last year, scaled to the recent level
Secondary sales	What distributors sell to retailers
Sell-in	The weeks when the trade stocks up ahead of a festival, such as the Durga Puja or Diwali sell-in
SHAP	A method for measuring how much each feature moves a prediction
SKU	A single product code; 30 in the data
Supply availability	The share of demand plants can ship in a scenario
Target month	The month a forecast run is for
Tertiary sales	What retailers sell to consumers; not in Eveready's data
Thursday rule	An earlier way of assigning a week to a month by its Thursday; now used only to pick which month's data a selected week shows
Trade scheme	A discount or incentive offered to the trade; barely used by the model (29th of 35 features, about +0.2 points of accuracy)
Transfer (inter-depot)	Moving stock from an overstocked depot to a short one; 503 historical transfers in the data
Under-forecast	A forecast below actual sales, a lost-sales risk
Value (₹)	Sales in rupees, units times the average selling price
Vintage	A forecast run made at one cut-off date, such as the end of a month
Weighted distribution	Share of category sales through shops that stock a product
Weekly outlook	The weekly forecast view on the Forecast screen

Known limits and open items

- **No new products screen.** The competitor file has 8 liquid vaporizer brands and several comparable launches, but no Eveready rows, so a vaporizer case would be a forecast with nothing to check it against. `competitor_data_v2.csv` adds Eveready but no script uses it. npd03 and npd04 show sales before their `launch_date`.
- **Product-level accuracy trails the sales team nationally.** The pooled total looks strong, but scored product by product the model is 6 to 8 points below the sales team. Festival peaks are still under-forecast, because each festival has only two or three repeats in the history. The agreed fix is one more year of history from Oct 2022 and a baseline-plus-festival-curve structure; neither is built.
- **Short test window.** Only ten completed months have M3, M2 and M1 runs.
- **Stale screen text.** The Weather screen still says the 2025 monsoon was neutralised, and the Quick commerce screen says ten products where the data has sixteen. Returns at 39% of delivered units look implausible.
- **Scenario controls** apply to the weekly outlook only, not to the monthly runs.
- **Depot level is Kolkata only,** about 7.4% of national primary sales. Lost sales count only orders that were placed.
- **Target level not confirmed.** Primary sales is a working choice; the CFO's workbook does not say which sales level he means, and no tertiary data exists.
- **Not built:** Overview, Supply & inventory, Channels & events and Guide.
- **Housekeeping:** `run_all.sh` is out of date, and some leftover files in the Mac project folder need delete permission.